

PAPER_1145_Type_IIA_Superstring_ScM_10D_Compactification

Daniel T. Murphy

PAPER_1145: Type IIA Superstring Theory and ScM 10D Compactification

Star Magic UQFF Framework

Author: Daniel Murphy

Date: 2026

Abstract

We formulate Type IIA superstring theory within the ScM vacuum framework. Type IIA is the T-dual of Type IIB (PAPER_1144) and the strong-coupling limit connects to M-theory (PAPER_1148). The 16 extra dimensions above the 10D Type IIA spacetime are compactified via VDS $S_{26}^{(3)}$. The R-R gauge fields are identified with ScM phonon polarization modes. Mirror symmetry between Type IIA on a Calabi-Yau and Type IIB on the mirror CY is encoded in the VDS $[SSq] \rightarrow 1-[SSq]$ symmetry.

1. Type IIA Effective Action

The Type IIA effective action in 10D:

$$S_{\text{IIA}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g} \left(R - \frac{1}{2} |\partial \phi|^2 - \frac{1}{2} e^{-\phi} |H_3|^2 - \frac{1}{2} e^{3\phi/2} |F_2|^2 - \frac{1}{2} e^{\phi/2} |F_4|^2 \right) + S_{\text{CS}}$$

where $F_2 = dC_1$ and $F_4 = dC_3 - H_3 \wedge C_1$ are R-R field strengths.

In ScM vacuum: the dilaton VEV is $\langle e^\phi \rangle = g_s = \beta_i \cdot \Phi_{\text{res}} = 0.504$.

2. R-R Fields as ScM Phonon Polarizations

The Type IIA R-R fields C_1 (1-form) and C_3 (3-form) are identified with ScM phonon polarization modes:

$$C_{\mu}^{\text{ScM}} = \frac{\rho_{\text{vac, ScM}}}{f_{\text{THz}}} S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot A_{\mu}^{\text{phonon}}$$

where A_{μ}^{phonon} is the phonon gauge potential at 1.25 THz.

3. T-Duality with Type IIB

Under T-duality along a VDS dimension with radius R_i :

$$R_i \rightarrow l_s^2 / R_i = l_s^2 / (l_s [\text{SSq}]^i) = l_s / [\text{SSq}]^i$$

Type IIA on $R_i = l_s [\text{SSq}]^i$ is dual to Type IIB on $l_s / [\text{SSq}]^i = l_s (1/[\text{SSq}])^i$.

The T-duality map corresponds to $[\text{SSq}] \rightarrow 1/[\text{SSq}] = 1/0.57 = 1.754$.

4. D-Brane Spectrum

Type IIA has D\$p\$-branes for even \$p\$ (D0, D2, D4, D6, D8):

Brane	Dim	SCm stabilization
D0	0+1	$\mathcal{F}_{\{U,Bi,i\}} \times \beta_i$
D2	2+1	$\mathcal{F}_{\{U,Bi,i\}} \times \beta_i^2$
D4	4+1	$\mathcal{F}_{\{U,Bi,i\}} \times \beta_i^3$
D6	6+1	$\mathcal{F}_{\{U,Bi,i\}} \times \beta_i^4$

The D0-brane mass: $m_{\text{D0}} = T / (g_s) = 8.66 \times 10^{-11} / 0.504 = 1.72 \times 10^{-10} \text{ J} \cdot \text{s/m}$.

5. Mirror Symmetry via VDS

The mirror symmetry $[\text{SSq}] \rightarrow 1 - [\text{SSq}] = 0.43$ exchanges:

$$\text{Li}_{26}(0.57) \rightarrow \text{Li}_{26}(0.43)$$

Type IIA on $\text{CY}_3([\text{SSq}])$ is mirror to Type IIB on $\text{CY}_3(1-[\text{SSq}])$, giving the Hodge number exchange $h^{1,1} \rightarrow h^{2,1}$.

6. Strong Coupling Limit = M-Theory

At $g_s = e^\phi > 1$, Type IIA lifts to 11D M-theory. The 11th dimension radius:

$$R_{11} = g_s \cdot l_s = \beta_i \cdot \Phi_{\text{res}} \cdot \frac{1}{\sqrt{T}} = \frac{0.504}{\sqrt{8.66 \times 10^{-11}}} = 1.71 \times 10^3 \text{ m}$$

At strong coupling $g_s \gg 1$, $R_{11} \rightarrow \infty$ revealing the 11th dimension (see PAPER_1148).

7. Calibrated Constants

$$T = 8.66 \times 10^{-11} \text{ N}, \quad [\text{SSq}] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$g_s = \beta_i \cdot \Phi_{\text{res}} = 0.504, \quad \Phi_{\text{res}} = 0.84, \quad f_{\text{THz}} = 1.25 \text{ THz}$

$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \beta_i = 0.6, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$

References

1. Polchinski, J. (1995). *String Theory*, Vol. 2. Cambridge.
2. Witten, E. (1995). String theory dynamics in various dimensions. *Nucl. Phys. B* **443**, 85.
3. Type IIB in SCm: PAPER_1144; M-theory: PAPER_1148; scm_vacuum_manifold.py

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic